Spark Cluster with Elasticsearch Inside

Oscar Castañeda-Villagrán
Universidad del Valle de Guatemala

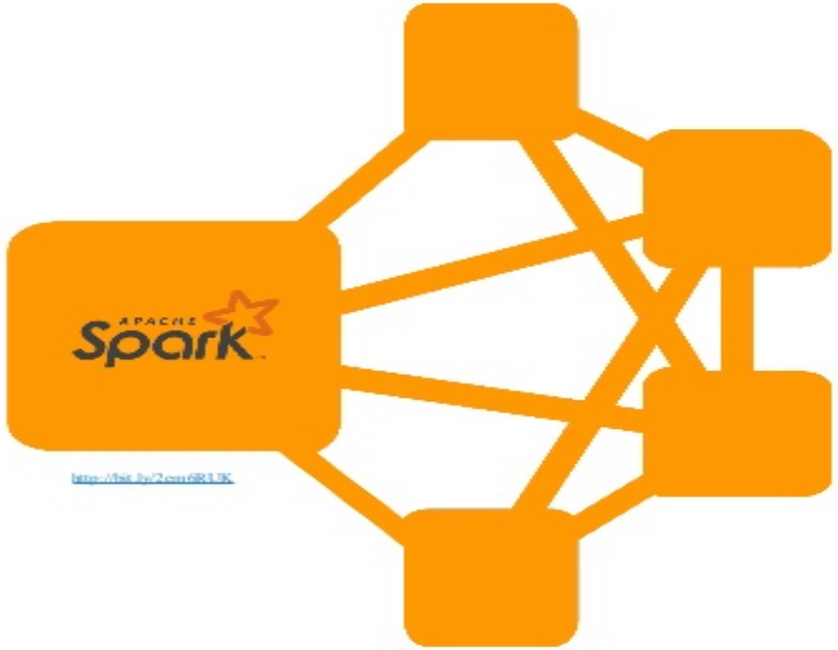
SPARK SUMMIT EUROPE 2016

About

- Researcher at Universidad del Valle de Guatemala.

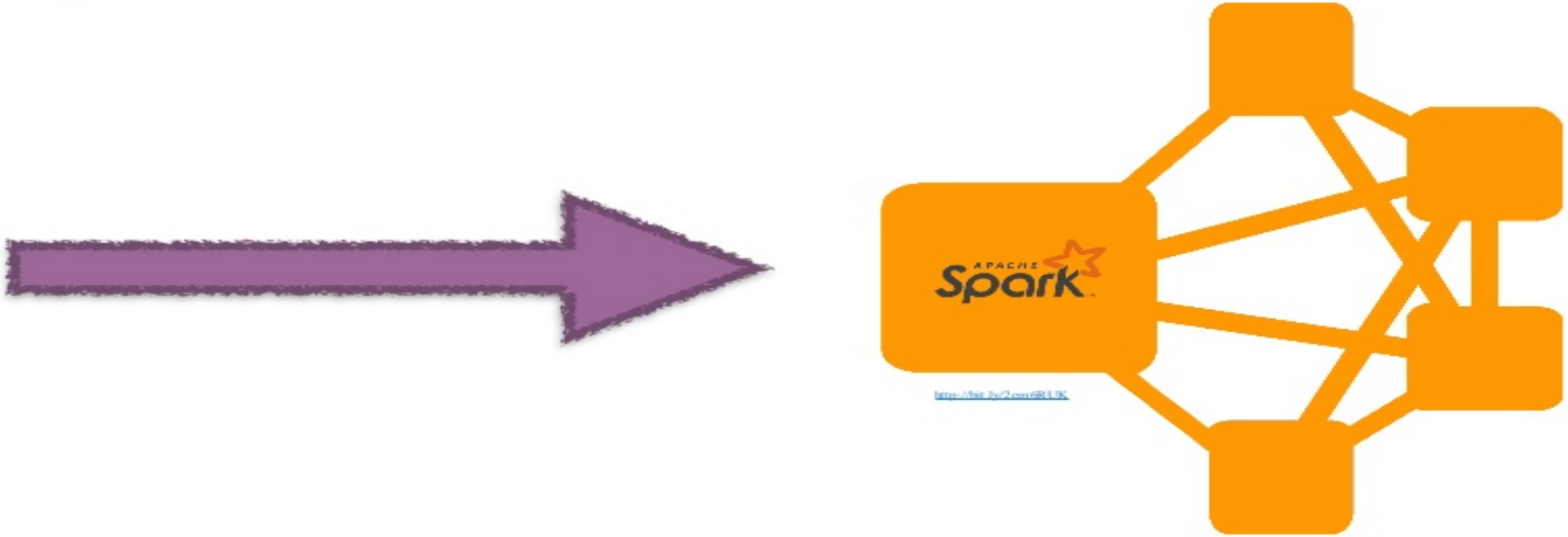
- Research Interests:
  - Program Transformation,
  - Programming Education Research,
  - Online Learning to Rank.

Spark cluster …



http://bit.ly/2cm6RUK

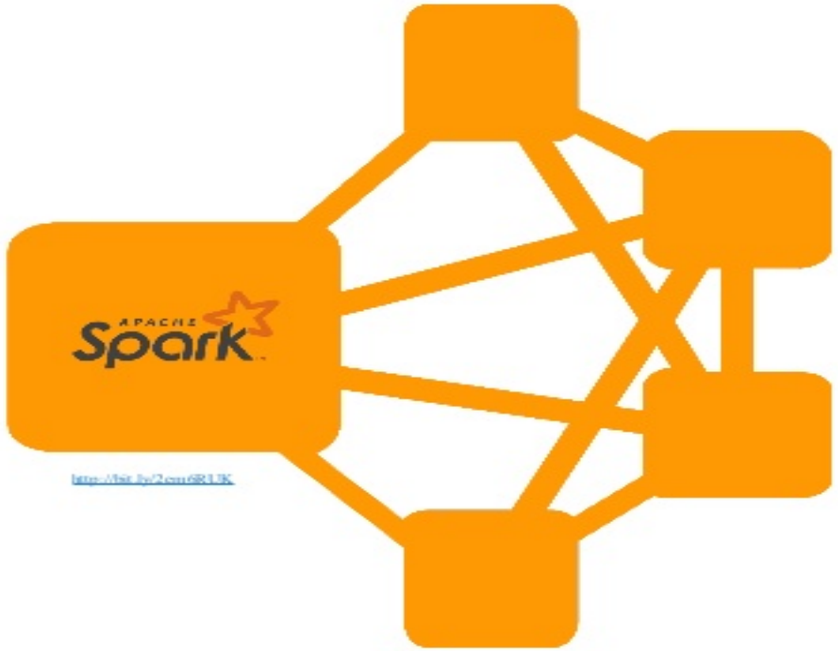
Spark cluster with …



Spark cluster with Elasticsearch

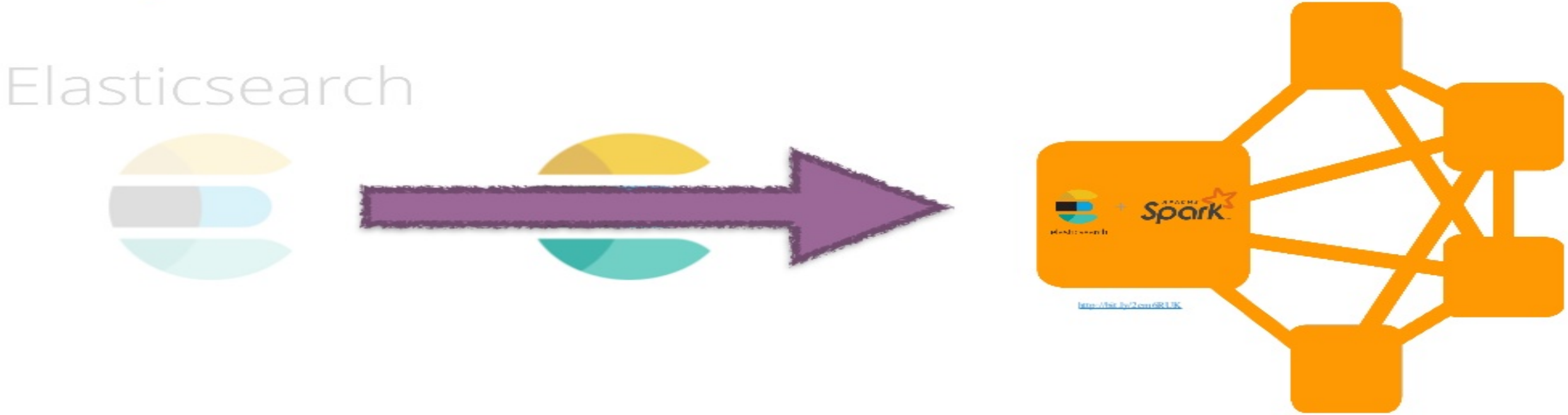
Elasticsearch

https://bit.ly/2eb9M9BO

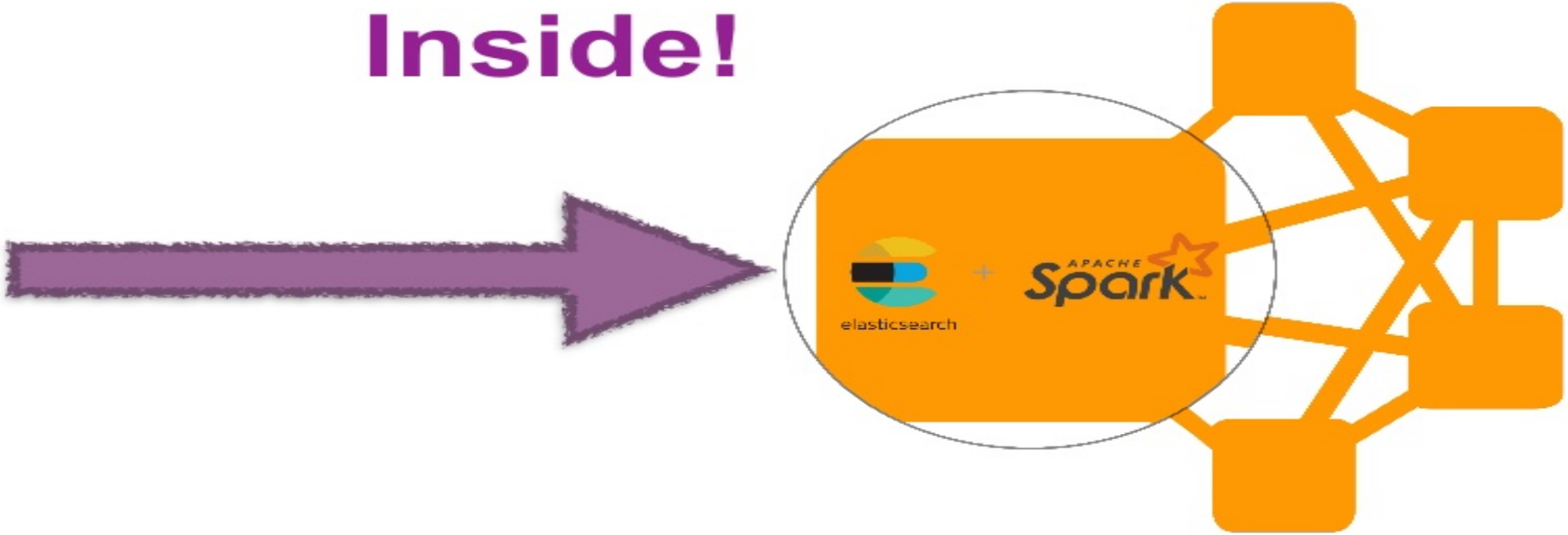


http://bit.ly/2eo6RUK

Spark cluster with Elasticsearch



Spark cluster with Elasticsearch Inside!



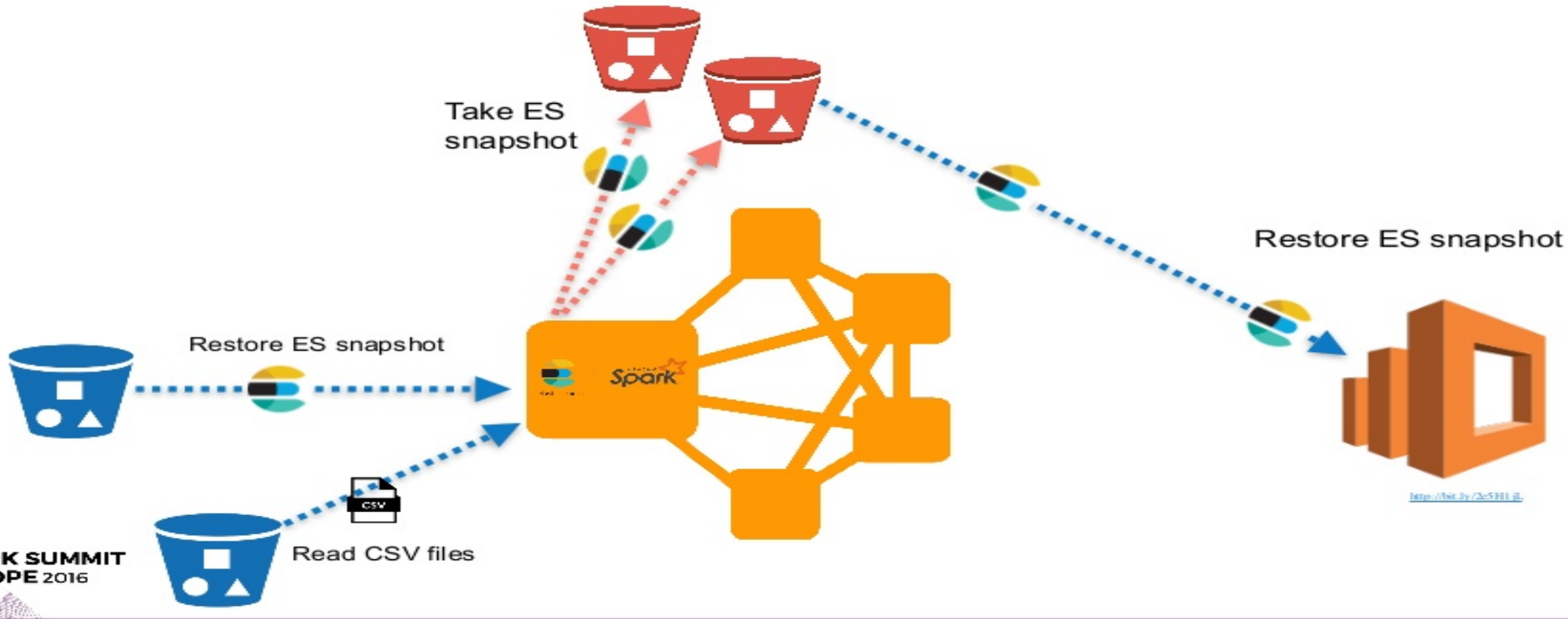
Agenda

- Problem Statement and Motivation.
- Read/Write (internal) ES Server.
- Create ES Server inside Spark Cluster.
- Snapshot/Restore ES indices using S3.
- Demo: IndexTweetsLive on Spark with Elastic inside.
- Q&A

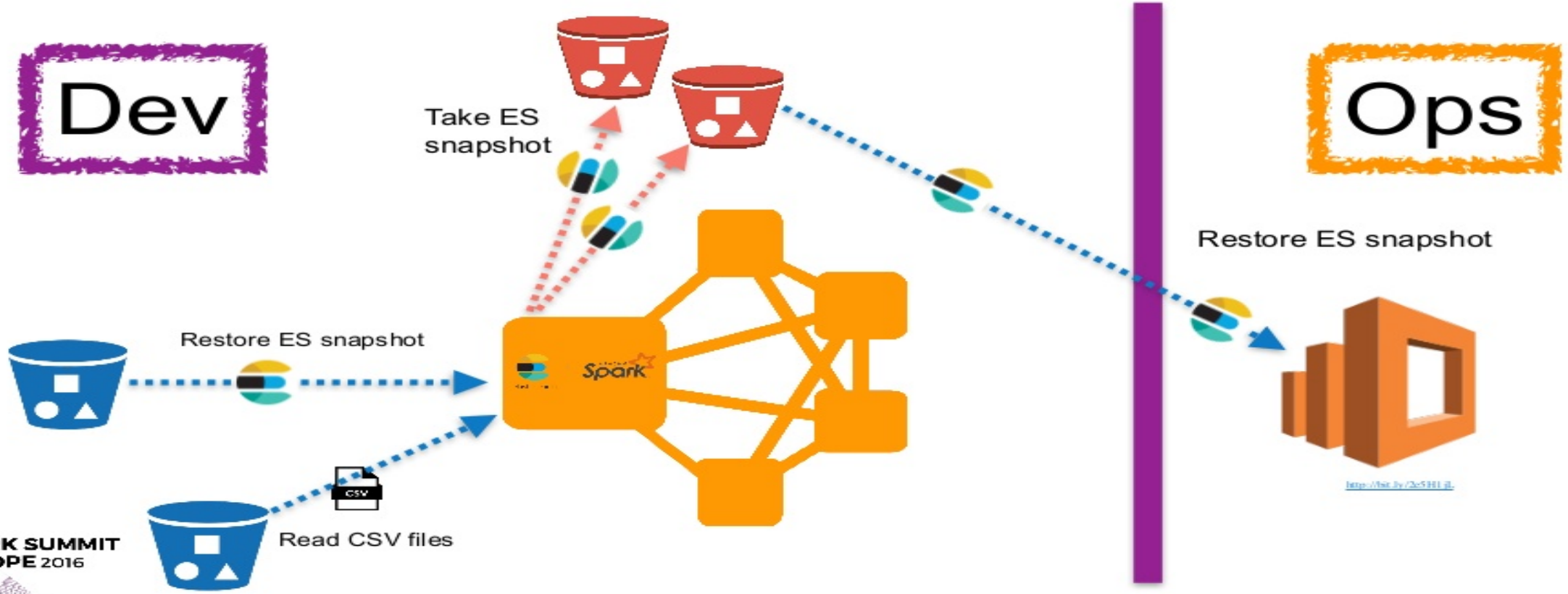
Problem Statement

- During development with ES-Hadoop it is cumbersome to have Elasticsearch running outside a Spark cluster.

Architecture

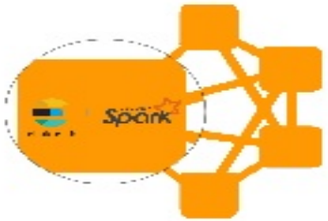


Architecture



Motivation

- Control Elasticsearch instance during development.
- Reduce dependencies between teams during development.
- Use ES snapshots as interface between teams.
- Increase QA efficiency.



Native Integration

Write data to Elasticsearch

```
import org.apache.spark.SparkContext
import org.apache.spark.SparkContext._

import org.elasticsearch.spark._

...

val conf = ...
val sc = new SparkContext(conf)

val numbers = Map("one" -> 1, "two" -> 2, "three" -> 3)
val airports = Map("arrival" -> "Otopeni", "SFO" -> "San Fran")

sc.makeRDD(Seq(numbers, airports)).saveToEs("spark/docs")
```

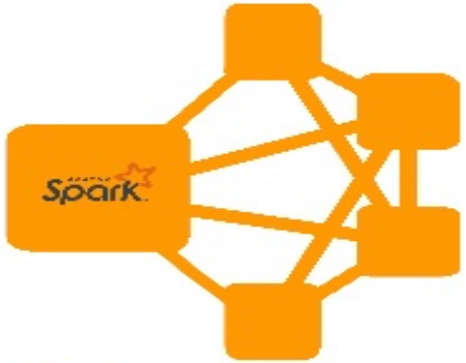

https://www.elastic.co/guide/en/elasticsearch/hadoop/current/spark.html#spark-write

Native Integration

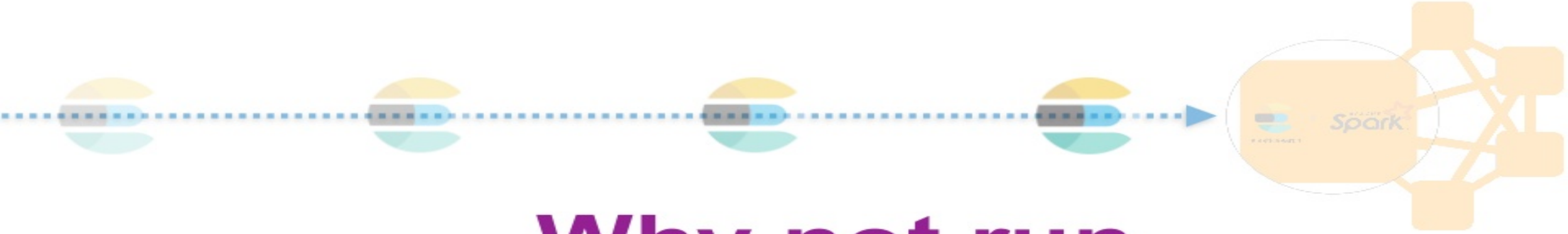
Read data from Elasticsearch

```
import org.apache.spark.SparkContext
import org.apache.spark.SparkContext._

import org.elasticsearch.spark._

...

val conf = ...
val sc = new SparkContext(conf)

val RDD = sc.esRDD("radio/artists")
```


https://www.elastic.co/guide/en/elasticsearch/hadoop/current/spark.html#spark-read

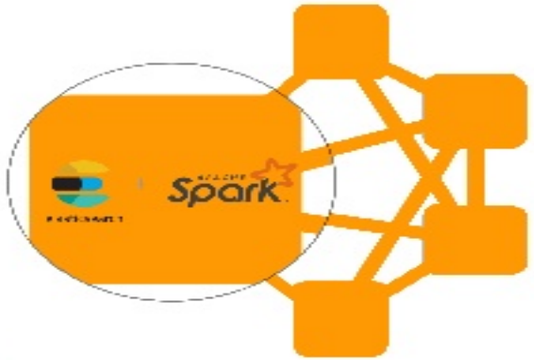


But where do you run Elasticsearch?



Why not run Elasticsearch inside Spark Cluster? *

* At least for development purposes.



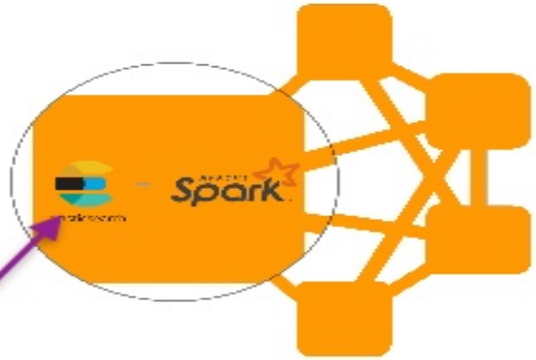
How do you run Elasticsearch inside Spark Cluster?

Imports

```
> import org.elasticsearch.client.Client
  import org.elasticsearch.common.settings.ImmutableSettings
  import org.elasticsearch.common.settings.Settings
  import org.elasticsearch.node.NodeBuilder._
  import org.elasticsearch.spark._
  import org.elasticsearch.action.get.GetResponse
  import org.elasticsearch.action.search.SearchResponse
  import org.elasticsearch.cloud.aws

  import org.apache.spark.SparkConf
  import org.apache.spark.SparkContext

  import com.sksamuel.elastic4s.{ElasticClient, ElasticsearchClientUri}
  import com.sksamuel.elastic4s.ElasticDsl._
  import com.sksamuel.elastic4s.mappings.FieldType._

  import scala.concurrent._
  import scala.concurrent.duration._
  import ExecutionContext.Implicits.global
```

Setup Local ES

```
> class ElasticsearchServer {
    private val clusterName = "sparksummiteu2016"
    private val settings = ImmutableSettings.settingsBuilder()
      .put("cluster.name", "sparksummiteu2016")
      .put("network.host", "10.104.239.70")
      .put("http.enabled", "true")
      .put("es.index.auto.create", "true")
      .build

    private lazy val node = nodeBuilder().local(false).settings(settings).build
    def client: Client = node.client

    def start(): Unit = {
      node.start()
    }
    def stop(): Unit = {
      node.close()
    }
  }

  val server = new ElasticsearchServer
  server.start()
```




```
server.start()
```

Write to Local ES

```
> val conf = new SparkConf().setAppName("sparksummiteu2016").setMaster("local[2]")
  conf.set("es.index.auto.create", "true")
  conf.set("spark.driver.allowMultipleContexts", "true")

  val sc2 = new SparkContext(conf)

  val ESAgents = sc2.parallelize(jsonAgents)

  ESAgents.saveToEs("tweets/hashtags")
```

```
saveToEs("tweets/hashtags")
```

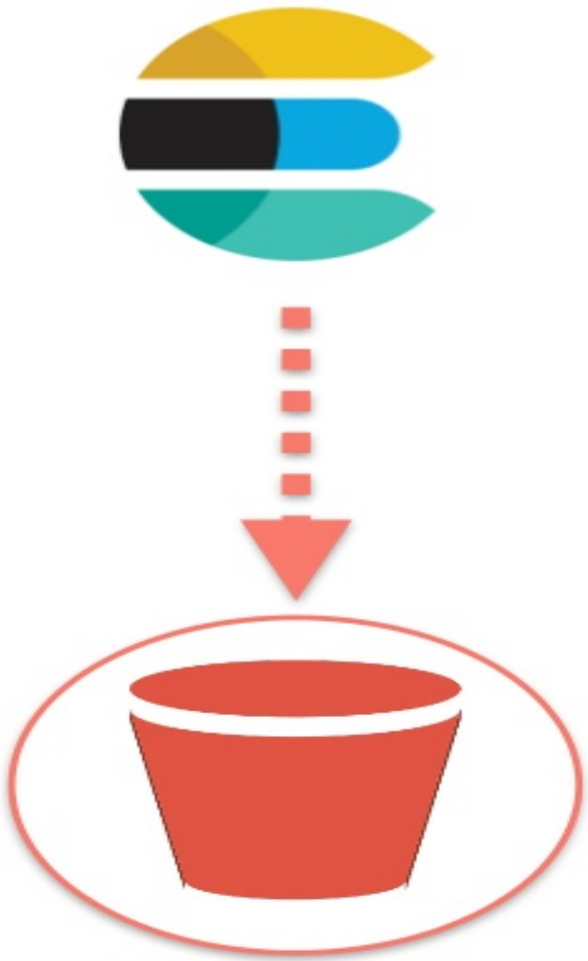
Check results on local ES

```
> def getUrlAsString(url: String): String = {
    val client = org.apache.http.impl.client.HttpClientBuilder.create().build()
    val request = new org.apache.http.client.methods.HttpGet(url)
    val response = client.execute(request)
    val handler = new org.apache.http.impl.client.BasicResponseHandler()
    handler.handleResponse(response).trim
  }

  getUrlAsString("http://10.104.239.70:9200/_cat/indices?v")
```

```
getUrlAsString("http://10.104.239.70:9200/_cat/indicies?v")
```

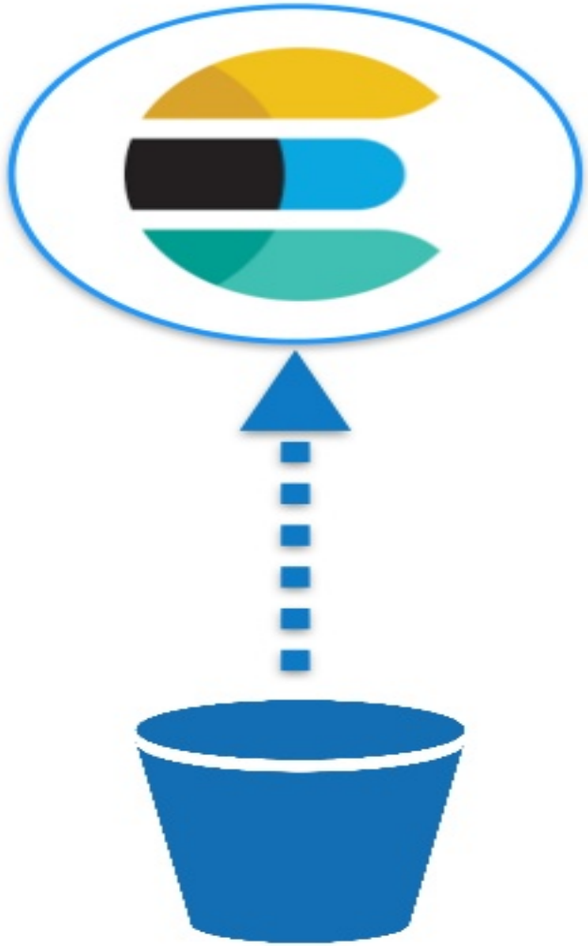
Snapshot to S3

```
> val uri = ElasticsearchClientUri("elasticsearch://10.104.239.70:9300")
  val settings = ImmutableSettings.settingsBuilder()
    .put("http.enabled", "true")
    .put("cluster.name", "elasticagents")
    .put("path.repo", "/your/path/")
    .put("access_key", "yOuRaCcEsSkEy")
    .put("secret_key", "yOuRsEcReTkEy")
    .build()

  val client = ElasticClient.remote(settings,(uri))

  val test = client.execute {
    create repository myRepo `type` "s3" settings Map("bucket" -> "bucket.name", "region" ->
  "us-west-1", "access_key" -> "yOuRaCcEsSkEy", "secret_key" -> "yOuRsEcReTkEy")
  }.await

  client.execute {
    create snapshot mySnapshot1 in myRepo waitForCompletion true
  }.await
```



Restore from S3

```
> val uri = ElasticsearchClientUri("elasticsearch://10.104.239.70:9300")
  val settings = ImmutableSettings.settingsBuilder()
    .put("http.enabled", "true")
    .put("cluster.name", "elasticagents")
    .put("path.repo", "/your/path/")
    .put("access_key", "yOuRaCcEsSkEy")
    .put("secret_key", "yOuRsEcReTkEy")
    .build()

  val client = ElasticClient.remote(settings,(uri))
  val repo = "myRepo"
  val snap = "mySnapshot1"

  client.execute { closeIndex("myIndex")}

  client.execute {
    create repository repo 'type' "s3" settings Map("bucket" -> "your.bucket", "region" -> "us-
  west-1", "access_key" -> "yOuRaCcEsSkEy", "secret_key" -> "yOuRsEcReTkEy")
  }.await

  val t = client.execute {
    restore snapshot mySnapshot1 from repo waitForCompletion true
  }
  Await.result(t, 1000 seconds)
```



Demo!

What have we seen?

- How to Read/Write (internal) ES Server.
- How to create ES Server inside Spark Cluster.
- How to Snapshot/Restore ES indices using S3.
- Demo: IndexTweetsLive on Spark with Elastic inside.

Next Steps

- Spark 2.0
- Continuous Applications
- Elasticsearch 5.0

Q&A

THANK YOU.

Email: ofcastaneda@uvg.edu.gt
Twitter: @oscar_castaneda

SPARK SUMMIT EUROPE 2016